Look Again: Self-Correcting Tool Use for Panoramic Dental Radiograph Diagnosis

Reza Nadimi* [Second author name] [Third author name]
[affiliations]

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Abstract

Panoramic dental diagnosis requires both identifying a finding and assigning it to the correct tooth. In our zero-shot evaluation on DENTEX, the strongest baseline reaches a pooled exact finding-level F1 of only 0.222, which motivates methods that can revisit localized image evidence. We present **VLM-DENTAL**, a tool-using vision-language pipeline built on one principle: *grounding is a hypothesis to interrogate, not an answer to trust.* An eight-tool interface lets a policy inspect native-resolution regions and revise a proposed box through box-relative shifts and scales (`nudge_crop`). During demonstration synthesis, tiered spatial perturbations expose the teacher to imperfect grounding without verbal error cues, and a ground-truth-anchored reward is designed to preserve corrective tool use during reinforcement learning. The released corpus contains 3,694 verified, decontaminated synthetic traces. On ground-truth-anchored pathology locates, the teacher issues corrective nudges on 38.4% of large offsets, 13.6% of small offsets, and 1.1% of clean boxes; these are demonstration statistics, not trained-policy results. We further specify a versioned rules-and-skills harness and a separately gated speculative-decoding benchmark, which keep inference-time procedural context distinct from model training and serving optimization. Evaluation follows a frozen protocol with DENTEX-native set matching and paired image-level bootstrap intervals; trained-policy, harness, and serving results are pending and are not claimed.

1 Introduction

Panoramic dental radiograph analysis requires assigning diagnostic findings to individual teeth in standard FDI notation. General-purpose vision-language models (VLMs) offer a flexible interface, but in our zero-shot evaluation on DENTEX they agree only weakly with expert annotations (Section 4.2); the best pooled exact F1 is 0.222. Image-level hit rates and calibration give complementary evidence but do not identify the causes of these errors. Local detail, tooth identity, and orientation all matter, which motivates an agent that can inspect a region again before answering.

Tool-using systems in medical and general domains provide relevant precedents (Wu et al., 2026; Li et al., 2024; Lu et al., 2025; Fan et al., 2026; Hao et al., 2026). We focus on one failure point: a learned grounding model can return a plausible box for the wrong region, and every subsequent crop inherits that error. Instead of treating the box as an authoritative observation, the agent should be able to inspect and revise it. Whether a trained agent acquires this behavior, and whether it improves diagnosis, are empirical questions that the frozen evaluation design is built to answer.

*Corresponding author. [Affiliations and co-author list to be finalized.]

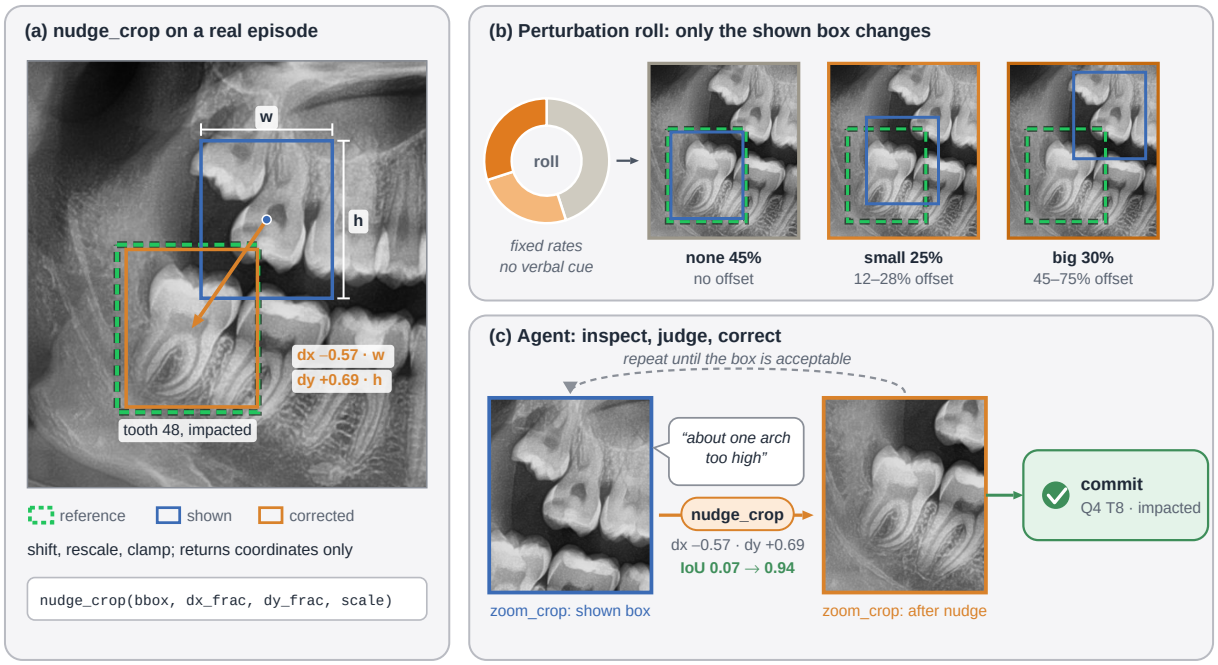


Figure 1: Grounding is a hypothesis, not an answer (DENTEX image 127, impacted FDI tooth 48). (a) `nudge_crop` shifts a wrong box (blue, 0.92 box-sizes from the expert box, green) onto the tooth. (b) The demonstrations were generated with a tiered perturbation of the shown box. (c) The agent inspects, judges ("about one arch too high"), corrects, and answers from the corrected view (IoU 0.07→0.94).

This paper develops VLM-DENTAL, an agentic VLM for OPG diagnosis built on a different contract between the agent and its tools: *grounding is a hypothesis, not an answer* (Fig. 1). Fig. 2 shows the system, Fig. 3 opens its model, and Fig. 7 contrasts the two trace arms on one image. Our contributions are:

1. **A self-correcting grounding tool (`nudge_crop`, Section 3.4).** The agent can shift and rescale a box it was *already given*, whether by the grounding model or by a previous correction, using box-relative fractional arguments (`dx_frac`, `dy_frac`, `scale`) that a VLM can reason about in words ("shift right about half a tooth-width"), without re-running detection. Box revision thus becomes an explicit, inspectable action rather than an implicit change in the model's internal localization.

2. **A tiered perturbation mechanism that makes self-correction learnable (Section 3.5).** Demonstrating correction behavior requires a demonstrator that *experiences* tool error. During trace generation we return the expert-verified box for annotated teeth, which keeps the task solvable, but perturb the box *shown to the demonstrator*: 45% clean, 25% small offsets (12–28% of box size), and 30% large offsets (45–75%). The rates are fixed constants decoupled from the detector's real error rate, so demonstrations do not go stale as the detector improves, and errors are never announced verbally, so correction has to be elicited by the returned image rather than by an explicit error message. Transfer of this behavior to the trained agent remains to be measured. A corpus-level analysis (Section 4.4) shows the mechanism elicits exactly the intended behavior gradient in 3,694 generated demonstrations: corrective use of `nudge_crop` on 38.4% of large offsets, 13.6% of small ones (intermediate offsets), and 1.1% of clean boxes, on ground-truth-anchored pathology locates.

3. **A tool interface with continuous parameters (Section 3.3).** Tools expose *how much* effect to apply, not only which tool to call: `denoise(strength` $\in [0, 1]$`)`, `window_level` with exact `center`/`width` values as an override on top of presets, a multiplicative `enhance_contrast` factor, and a tunable `zoom_crop` context margin. Tool calls therefore carry information beyond tool identity, and the agent can match processing intensity to image characteristics.

4. **A ground-truth-anchored, correction-friendly reward (Section 3.8).** GRPO training (DeepSeek-AI, 2025) uses a composite reward whose dominant term is a hard, non-LLM-judge match against expert labels with multi-finding greedy one-to-one matching (F1-style harmonic mean of recall and precision). Its efficiency term scales a reference budget with case complexity and *exempts* `locate_tooth` and `nudge_crop` from per-call cost, so the reward does not penalize the verification behavior it is meant to teach.

5. **Label-anchored synthetic demonstrations and an audit trail (Section 3.6).** 3,694 verified chain-of-thought tool-use traces over DENTEX and Tufts (1,847 with tools, 1,847 single-turn without, exact 1:1 parity), generated by frontier models under real tool execution in a LangGraph loop, verified cross-family, decontaminated of directive leakage (105 traces regenerated; 0 leaks in the canonical corpus), and released as a reusable resource. The verifier's rejection taxonomy (66 false-positive findings on healthy scans, 15 ground-truth contradictions, 11 uncorrected spatial drifts) also serves as an audit of what the perturbation mechanism does and does not elicit.

We evaluate on DENTEX with its native structured metrics (pooled, set-matched, finding-level precision, recall, and F1 with paired bootstrap intervals) against zero-shot frontier VLMs, the same backbone with and without tools, and a supervised specialist detector. The corpus-level analyses reported here are complete; results of the two-stage training (SFT, then GRPO) are pending and are marked as such in the tables rather than extrapolated. Inference-time rules, skills, and serving optimizations are evaluated separately on the training-derived dev split (Section 5) and do not revise the frozen primary protocol.

2 Related Work

Supervised dental AI. Dental radiograph analysis has been dominated by narrow, single-task supervised models: CNN and Faster R-CNN cephalometric landmark detectors, U-Net-style segmentation, FDI tooth-numbering classifiers, and YOLO-family or Faster R-CNN detectors for caries, periapical lesions, and impactions (Hamamci et al., 2024). The DENTEX benchmark consolidated these into a hierarchical quadrant–enumeration–diagnosis task with expert-verified labels (Hamamci et al., 2024) and remains the most structured public benchmark for our problem. These supervised tasks differ from our episode-level setting, in which a policy can request additional views and revise a grounding hypothesis. A systematic review of public dental imaging datasets (Uribe et al., 2024) screened 131,028 records and found only 16 datasets that met a minimal bar, which motivates our synthetic-trace approach.

Reinforcement-learned reasoning in medical VLMs. Following the GRPO recipe (DeepSeek-AI, 2025), Hu et al. (2025) fine-tuned a 2B VLM with GRPO on a few hundred examples and reported large gains over zero-shot across MRI, CT, and X-ray VQA. Med-R1 extended this to eight modalities and found that longer chains of thought do not reliably improve accuracy, so reward alignment matters more than reasoning length (Lai et al., 2026). HuatuoGPT-o1 (Chen et al., 2024) and OpenMedReason (Baghbanzadeh et al., 2026) build reasoning supervision by using a frontier

model as a *verifier* of evidence-grounded traces. These works motivate our training recipe; we add the interaction between a reasoning policy and revisable visual observations.

Agentic tool use in VLMs. MMedAgent trains an LLM to select among grounding, segmentation, and report tools (Li et al., 2024), and OctoTools generalizes the orchestration pattern (Lu et al., 2025). Closest in spirit is VTool-R1 (Wu et al., 2026), which trains a VLM with GRPO to call Python-based visual-editing tools during reasoning and improves chart and table QA from 64.7% to 71.7%. Its tools are generic image-editing primitives, its reward is a single outcome-accuracy term, and its domain is not medical. Two concurrent dental systems are closest to our work. **OralGPT-Plus** (Fan et al., 2026) combines trained chain-of-thought, two callable image tools (zoom-in and contralateral mirror), and GRPO-style RL with a composite reward. It evaluates on its own benchmark with an LLM-judge-scored reward, whose documented reward-hacking failure motivates our ground-truth-anchored accuracy term (Section 3.8), and it does not cleanly separate the effects of tool use and RL. **OralAgent** (Hao et al., 2026) is a broader prompted ReAct orchestrator around an off-the-shelf model, not an RL-trained tool-use policy. Our contribution is a box-relative corrective action and a controlled evaluation of tools and RL; we do not claim to originate dental agents or medical tool use. Comparisons should account for differences in datasets, available supervision, and scoring protocols.

Procedural context and speculative serving. The rules-and-skills extension follows the inspectable `SKILL.md` convention with progressive disclosure (Agent Skills contributors, 2026); it contributes a provenance-tracked integration with the shared dental dispatcher, not a new knowledge-file format. Speculative decoding uses a draft proposal and target verification to accelerate generation (Leviathan et al., 2023; Chen et al., 2023), and vLLM provides the serving substrate (Kwon et al., 2023). We treat serving speed and clinical-task performance as separate measurements and test output equivalence on the actual multimodal episodes instead of assuming it from the algorithm.

Label-anchored synthetic reasoning traces. Frontier LLMs can generate and verify reasoning traces of sufficient quality for downstream fine-tuning (Chen et al., 2024; Baghbanzadeh et al., 2026), replacing prospective expert annotation with expert-*verified benchmark labels* plus cross-model automated checks. We extend this pattern to *multi-turn tool-use* traces: the demonstrator operates a real tool suite under real tool feedback, including deliberately imperfect tool output (Section 3.5), and an independent cross-family verifier audits every trace against expert labels before it enters the training set.

3 VLM-DENTAL: System and Method

3.1 Problem setup and agent loop

An OPG I is annotated as a *set* of findings $F = \{f_1, \ldots, f_m\}$ (possibly empty for a healthy image), where each finding $f = (\text{quadrant}, \text{tooth position}, \text{diagnosis})$ uses FDI two-digit notation over DENTEX's four diagnosis classes (caries, deep caries, periapical lesion, impacted tooth). The agent receives the image and a registry of tools, and produces a sequence of turns; each turn contains free-form clinical reasoning and zero or more structured tool calls, and the loop ends with a structured final answer reporting *every* finding the agent commits to. The same agent loop is used for synthetic data generation, RL rollouts, and evaluation, and tool calls execute for real against the source image at every stage (Fig. 2); a single shared dispatcher defines which tools consume the

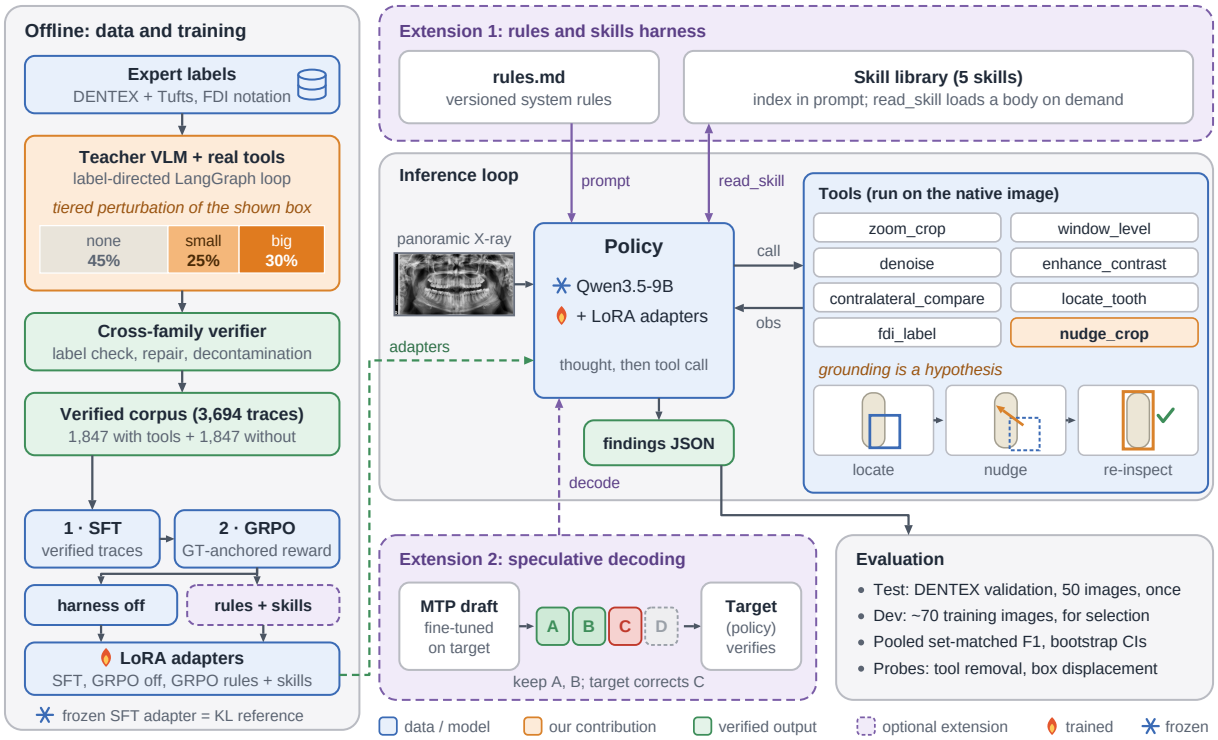


Figure 2: System overview: offline data and training (left), the agent loop with its eight tools (centre), and two optional extensions, a rules-and-skills harness and speculative decoding. Tools always run on the native image.

image and which image they operate on, so the training and inference environments cannot drift apart. Tool calls are capped at 50 per episode, and a lightweight JSON extractor repairs truncated or markdown-wrapped outputs.

3.2 System architecture

Fig. 2 shows the complete system: the agent loop used at inference and evaluation time, and the offline stack that produces its weights. Five component groups surround one agent core — a Qwen3.5-9B vision-language model running the loop of Section 3.1: a data layer (Fig. 4: registry-driven loaders for the two in-use sources, DENTEX and Tufts; a candidate pool of eight further public panoramic datasets, promoted only after their annotation semantics are verified), the trace synthesis stack of Section 3.6, the two training stages of Section 3.8, and an evaluation layer (Section 4.1) whose probes feed model selection on a fixed dev split.

Fig. 3 opens the model. Qwen3.5-9B has a 27-block vision transformer (frozen), a two-layer merger (trained), and 32 text layers: 24 Gated DeltaNet layers and 8 full-attention layers in a 3:1 pattern, each followed by an MLP, over a 4-bit NF4 base (QLoRA). Training adds LoRA to the attention, DeltaNet and MLP projections; the optional speculative-decoding extension (Section 5.3) also fine-tunes the native multi-token-prediction (MTP) block. Embeddings, norms, convolutions and `lm_head` stay frozen.

Three single-source-of-truth decisions hold the system together. First, one shared dispatcher defines, for every tool, whether and *which* image it consumes — always the native source image — so the trace-generation loop, the GRPO rollout loop, and the evaluation runner cannot silently diverge. Second, the image-input geometry lives in one module and is an explicit flag, not an implicit

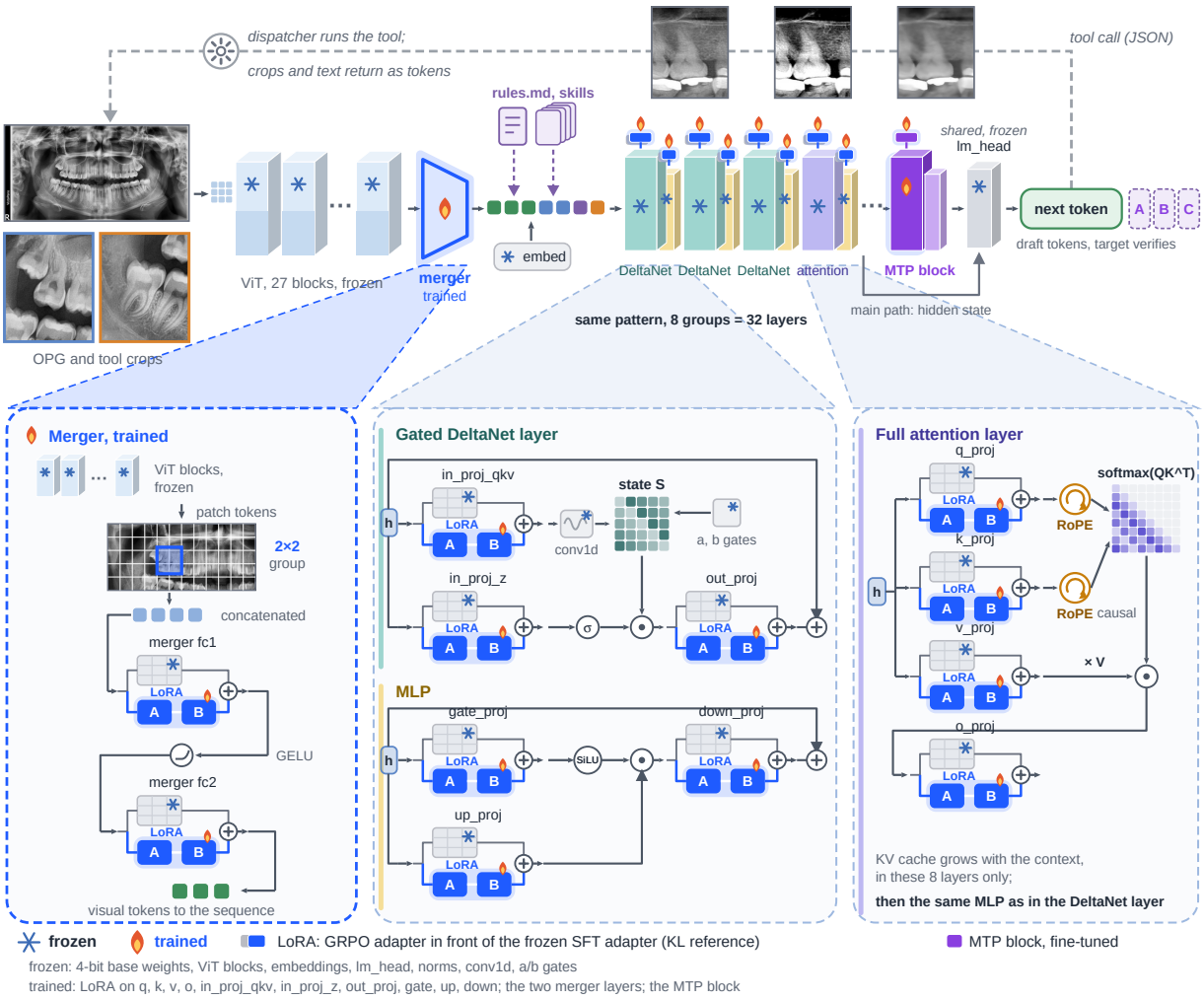


Figure 3: The model, Qwen3.5-9B, opened. Frozen parts are pale with a snowflake; trained parts are blue with a flame: LoRA on the attention, DeltaNet and MLP projections, the two merger layers, and the MTP block.

behavior. Training and evaluation run on GPU with the original images and dynamic shapes, identically across SFT, RL, and evaluation, and every tool operates on native-resolution pixels, so coordinates the model was trained to emit remain valid at inference. (The module also implements aspect-preserving canonical views with static vision-slot padding for static-shape accelerators such as TPUs; this path is not used in this work.) Third, the conversion from DENTEX's zero-indexed annotation rows to FDI notation has exactly one implementation, called by the reward, the verifier, and the metrics alike — so an indexing error cannot arise independently in several subsystems.

3.3 A tool suite with continuous parameters

The registry exposes eight tools (Table 1; Fig. 5 shows what they return). The deliberate design choice throughout is that tools expose *how much* effect to apply as a continuous or overridable parameter, not just a fixed on/off call: denoising takes a continuous `strength`, windowing takes exact `center`/`width` values as an override on top of tissue presets, contrast takes a multiplicative `factor`, and cropping takes a tunable context margin. An agent calling `denoise(strength=0.3)` rather than `denoise(strength=0.9)` is expressing a patient-specific judgment about how grainy a

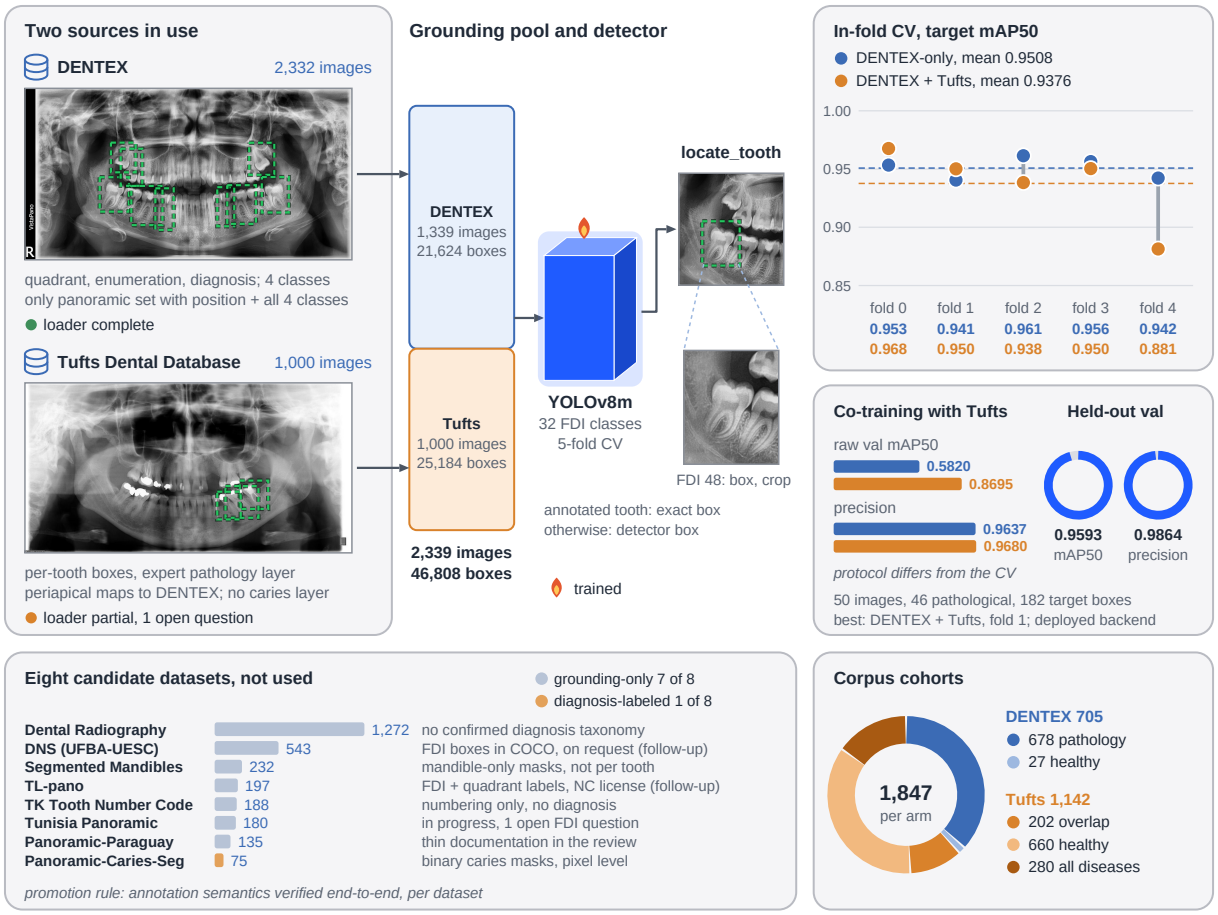


Figure 4: Data and the grounding detector: two real sources feed one grounding pool and the YOLOv8m detector behind `locate_tooth` (five-fold validation, held-out result); eight candidate datasets are not used; the corpus cohorts are shown as a donut.

crop is. A fixed three-level menu cannot carry this information, and the resulting tool-call stream is more informative to train on and to audit.

3.4 Self-correcting grounding: `nudge_crop`

The grounding tool is a YOLOv8m detector fine-tuned on 2,339 DENTEX+Tufts images with 46,808 tooth boxes, trained with 5-fold cross-validation (one class per FDI quadrant–position pair) (Fig. 4; full per-fold numbers in Appendix C). The detector is accurate (held-out *target* mAP50 of 0.93–0.96 with precision up to 0.986 on annotated teeth), but it is a 32-class detector with per-fold variance, and its errors propagate into downstream reasoning: a box shifted by half a tooth-width yields a crop that still *looks* like a plausible anatomical region while showing the wrong tooth.

`nudge_crop` makes such errors correctable. It takes a box the agent was *already given* (by `locate_tooth` or a previous `nudge_crop`; it cannot create a box from scratch) and returns adjusted coordinates: a shift (`dx_frac`, `dy_frac`) expressed as fractions of the box's own width and height, and a rescale `scale` around the new center, clamped to image bounds. The tool is deliberately *data-only*: it computes coordinates, and the agent must pair it with `zoom_crop` to actually see the corrected region. Two consequences follow. First, a fractional, box-relative parameterization ("shift left by about half the box") is easier for a VLM to reason about than raw pixel coordinates at an

Table 1: The VLM-DENTAL tool suite. Continuous or overridable parameters are shown in *italics*; data-only tools return coordinates rather than images.

Tool	Key parameters	Purpose
`zoom_crop`	bbox, *padding_frac*	High-resolution crop of a region with a tunable context margin
`window_level`	preset; *center*, *width*	Radiological intensity windowing (bone/enamel/soft-tissue), exactly overridable
`denoise`	method, *strength* $\in [0, 1]$	Edge-preserving bilateral or median filtering matched to crop noise
`enhance_contrast`	*factor*	Multiplicative contrast adjustment for subtle lesions
`contralateral_compare`	bbox, quadrant	Side-by-side mirror crop, constrained to the correct jaw half, for bilateral symmetry assessment
`locate_tooth`	FDI tooth	YOLOv8m grounding (32 classes, one per FDI position); returns a box
`fdi_label`	quadrant, position	FDI notation conversion and validation
`nudge_crop`	bbox, *dx_frac*, *dy_frac*, *scale*	Correct a box the agent was already given, without re-running detection

arbitrary zoom level. Second, because each correction must be followed by a verification crop, the loop enforces an inspect–correct–re-inspect cycle rather than a single blind adjustment. To our knowledge, neither concurrent dental agentic system exposes such a mechanism: their tools are fixed operations whose outputs the agent can only accept.

3.5 Tiered perturbation: making self-correction learnable

An agent learns to question its tools only if its demonstrations do. An idealized demonstrator, whose grounding tool always returns the right box, demonstrates *trusting* the tool, and SFT on such traces teaches the wrong reflex. At the same time, unguided exploration by the demonstrator would sharply reduce the yield of usable traces.

Our resolution (Fig. 1) separates what the grounding tool *computes* from what the demonstrator *sees*. During trace generation, `locate_tooth` operates in a ground-truth-grounding mode: for an annotated tooth it returns the expert-verified box (guaranteeing the demonstration remains solvable), while for exploratory calls on unannotated teeth it falls back to real detector inference. Then, independently of how the box was obtained, a *perturbation roll* decides what is displayed (exact configuration in Appendix A):

- **none** (45%): the box is shown as computed, giving a clean accept-and-proceed demonstration;
- **small** (25%): a synthetic offset of 12–28% of the box's own size per axis, random sign. This is a judgment call that the zoom margin often absorbs, and the prompt does *not* say whether such a box needs correction;
- **big** (30%): an offset of 45–75% per axis, sized to be visually self-evident.

Offsets are applied with `nudge_crop`'s own shift-and-clamp math, so every perturbed box is one the correction tool could plausibly have produced or fixed; the unperturbed box is logged internally for auditing but never shown. Two design choices matter more than the specific rates:

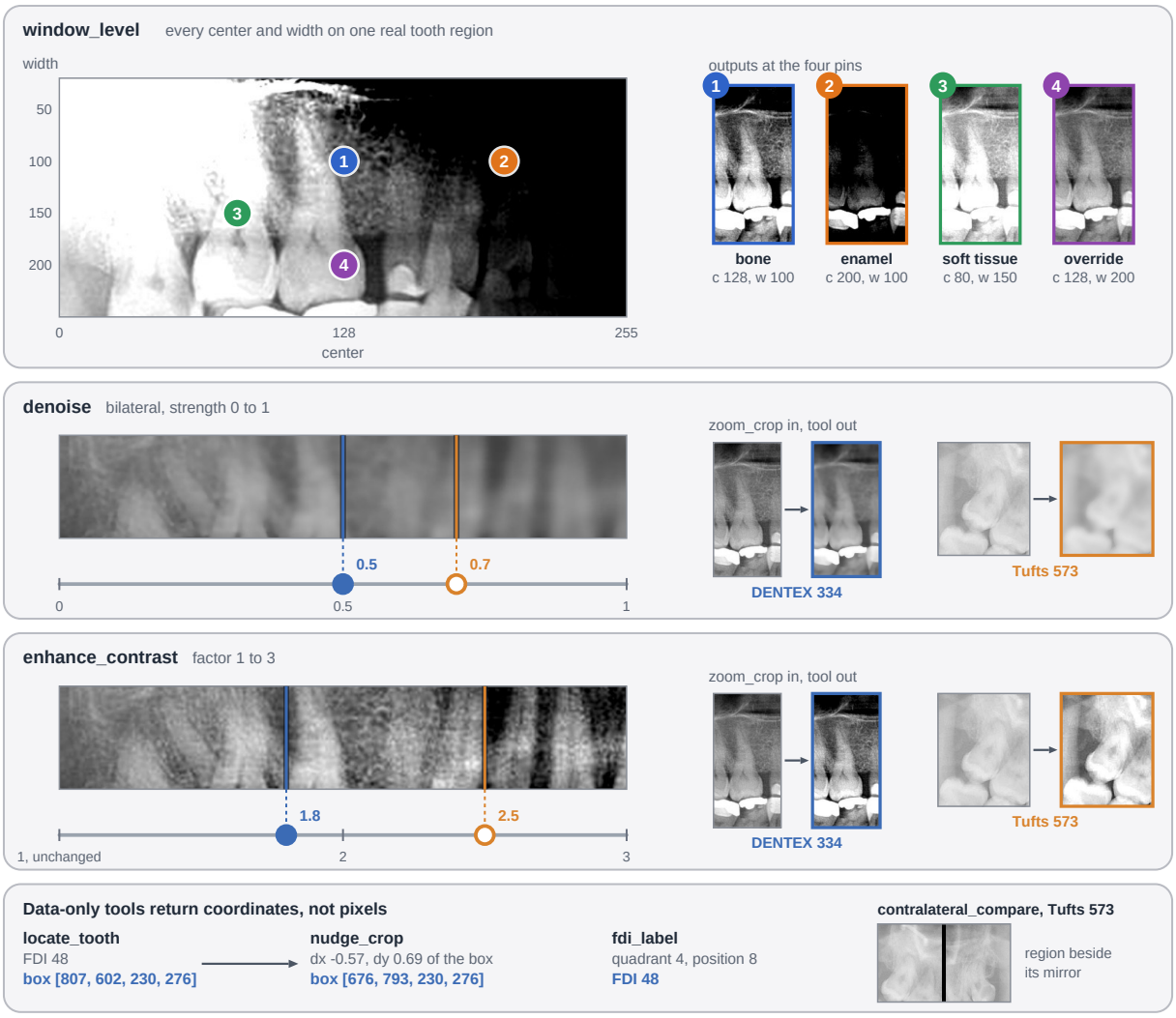


Figure 5: What the tools return, on one DENTEX region (image 334) and one Tufts region (image 573). Each sweep runs a tool across its full parameter range; pins mark the values the agent chose. Crops are native-resolution views with 30% padding.

Fixed rates, decoupled from detector accuracy. The perturbation probabilities are configured constants, not a reflection of the grounding model's measured error rate. This decoupling is deliberate: if the required frequency of verification tracked the detector's live precision, then every improvement to the detector would silently reduce the frequency of demonstrated correction, and traces generated under the old detector would teach a stale behavior. With fixed rates, the corpus remains valid as the grounding tool improves, as occurred during this project, when the detector's validated mAP50 improved from 0.58 (DENTEX-only) to 0.87–0.96 (multi-dataset) while the trace corpus remained in use unchanged.

No verbal cue; magnitude carries the signal. Nothing in the trace-generation prompt marks a perturbed box as wrong. A verbal cue ("this box needs correction") would exist only during data generation and not at RL or inference time, so teaching the model to respond to it would install a shortcut that fails at inference time. The big tier is therefore sized to be self-evident from the crop

alone, and the small tier is left ambiguous: the demonstrator has to look and judge, which is the skill the deployed agent must exercise without any ground truth.

3.6 Trace generation, verification, and decontamination

Traces are generated by a frontier teacher (Gemini 3.5 Flash Lite or MiniMax M3) that is directed toward the expert labels for the image but writes its own reasoning and tool calls turn by turn under real tool feedback. Generation is ground-truth-directed by design: unguided exploration by an untuned model would sharply reduce the yield of correctly labeled demonstrations. Every trace then passes an independent, cross-family *verifier* conditioned on the image and the expert labels, which rejects reasoning not supported by what the model was actually shown (rejection taxonomy in Appendix D). Verification is staged rather than monolithic (Fig. 6): a strict pass checks diagnostic correctness, FDI validity, and whether every claim is supported by what the model was actually shown — treating valid `nudge_crop` correction loops as evidence of diligence rather than as errors — and rejected traces are routed to a verifier-editor that repairs FDI labels, diagnosis labels, and formatting while preserving the clinical chain of thought; the repaired trace is re-verified before promotion. Both stages are cross-family by construction: the generator and the verifier are drawn from separate provider pools spanning different model families, so a generator's blind spot is unlikely to be shared by its judge. Because generation is conditioned on privileged information, traces are scanned for *directive leakage* — the demonstrator echoing its hidden instructions in its reasoning — and leaking cases are excised and regenerated under a dual gate (zero-leak regex audit + verifier re-approval); the canonical corpus contains zero leaks.

The resulting corpus (Table 2) comprises 3,694 verified traces over 1,847 images from two institutions: 1,847 interactive with-tools traces and 1,847 single-turn no-tools traces in exact 1:1 parity — the paired design that lets the evaluation isolate the contribution of the tools (Section 4.1). The with-tools corpus carries 5,144 finding-level ground-truth annotations across cohorts (4,513 unique, because the 202-image Tufts overlap cohort is contained in the 280-image Tufts all-diseases cohort), spanning caries (2,189), deep caries (578), impacted teeth (604), periapical lesions (789 unique), and three further Tufts lesion classes (353); multi-finding images carry up to 23 annotated findings, and the teacher surveys every finding on the image. Healthy scans are included as negative controls with empty finding sets. The verifier rejected 92 traces across three failure modes: 66 hallucinated findings on healthy scans, 15 ground-truth contradictions (wrong diagnosis class or swapped quadrants), and 11 uncorrected spatial drifts — cases where the teacher diagnosed from a visibly displaced box without issuing a corrective `nudge_crop`. The third category indicates that the perturbation mechanism creates judgment pressure that the verifier can audit. Across the with-tools corpus the teacher averaged 12–13 turns and 18–30 real tool calls per trace (Table 2); the full call distribution is shown in Fig. 7 (`nudge_crop` accounts for 2,292 calls).

3.7 Anatomy of a trace: with tools vs. without

Fig. 7 shows both halves of the paired corpus for one DENTEX image (a carious FDI tooth 34, a single finding). The no-tools trace is one turn: a fluent, well-structured direct diagnosis that reaches the right answer but leaves no evidence trail: its reasoning cannot be checked against anything the model was actually shown, and the model never had the opportunity to be wrong and recover. The with-tools trace for the same image runs 23 turns and 22 real tool calls and demonstrates the full behavioral repertoire the perturbation mechanism is designed to elicit: the grounding box for tooth 34 was inspected, judged wrong ("the crop shows only bone/root area … the box appears to be too low"), and corrected three times with box-relative `nudge_crop` shifts until re-inspection confirmed

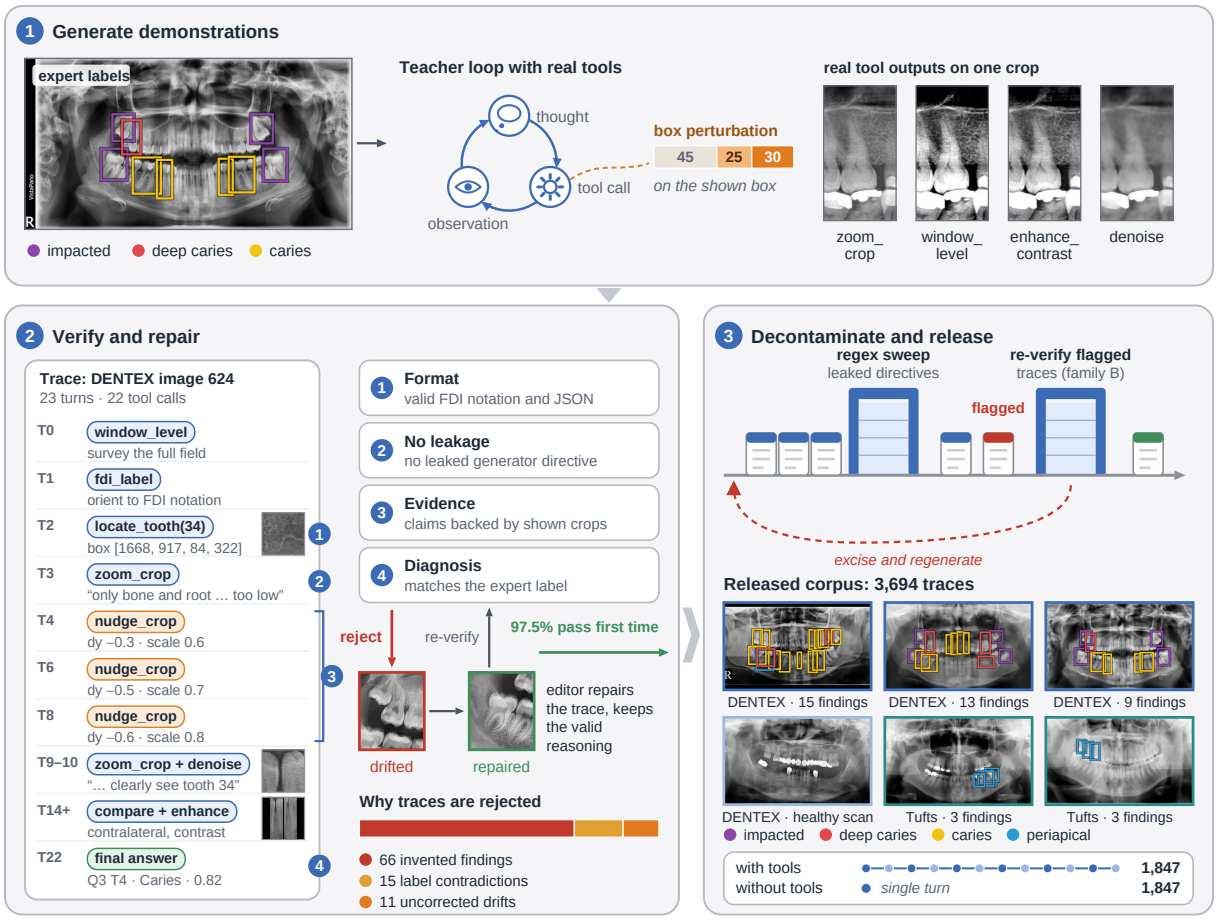


Figure 6: Trace synthesis, illustrated on one real trace (DENTEX image 624). (1) A label-directed teacher operates the real tools while the shown box is perturbed. (2) A cross-family verifier checks every trace; failures are repaired and re-verified. (3) A leak sweep excises and regenerates contaminated traces before release.

Table 2: The verified corpus. With-tools and no-tools cohorts are in exact 1:1 parity (checkmarks). Turns and tool calls are for the with-tools side; findings are counted per cohort (the Tufts overlap cohort is a subset of the Tufts all-diseases cohort, so its findings are counted in both).

Cohort	With-tools	No-tools	Findings	Healthy	Turns	Tool calls
DENTEX pathology	678	678 ✓	3529	0	12.9	25.2
DENTEX healthy (negative controls)	27	27 ✓	0	27	13.4	29.9
Tufts overlap pathology	202	202 ✓	631	0	12.0	18.0
Tufts healthy (negative controls)	660	660 ✓	0	660	12.8	27.0
Tufts 4-disease pathology	280	280 ✓	984	0	11.7	17.8
Total	1847	1847 ✓	5144	687	–	–

the crown; a contralateral symmetry comparison and a contrast-enhanced re-inspection followed before the structured final answer, which matches the expert label.

The paired design is what makes this contrast usable rather than anecdotal. With 1,847 with-tools and 1,847 no-tools traces in exact per-cohort parity, the evaluation can hold the image

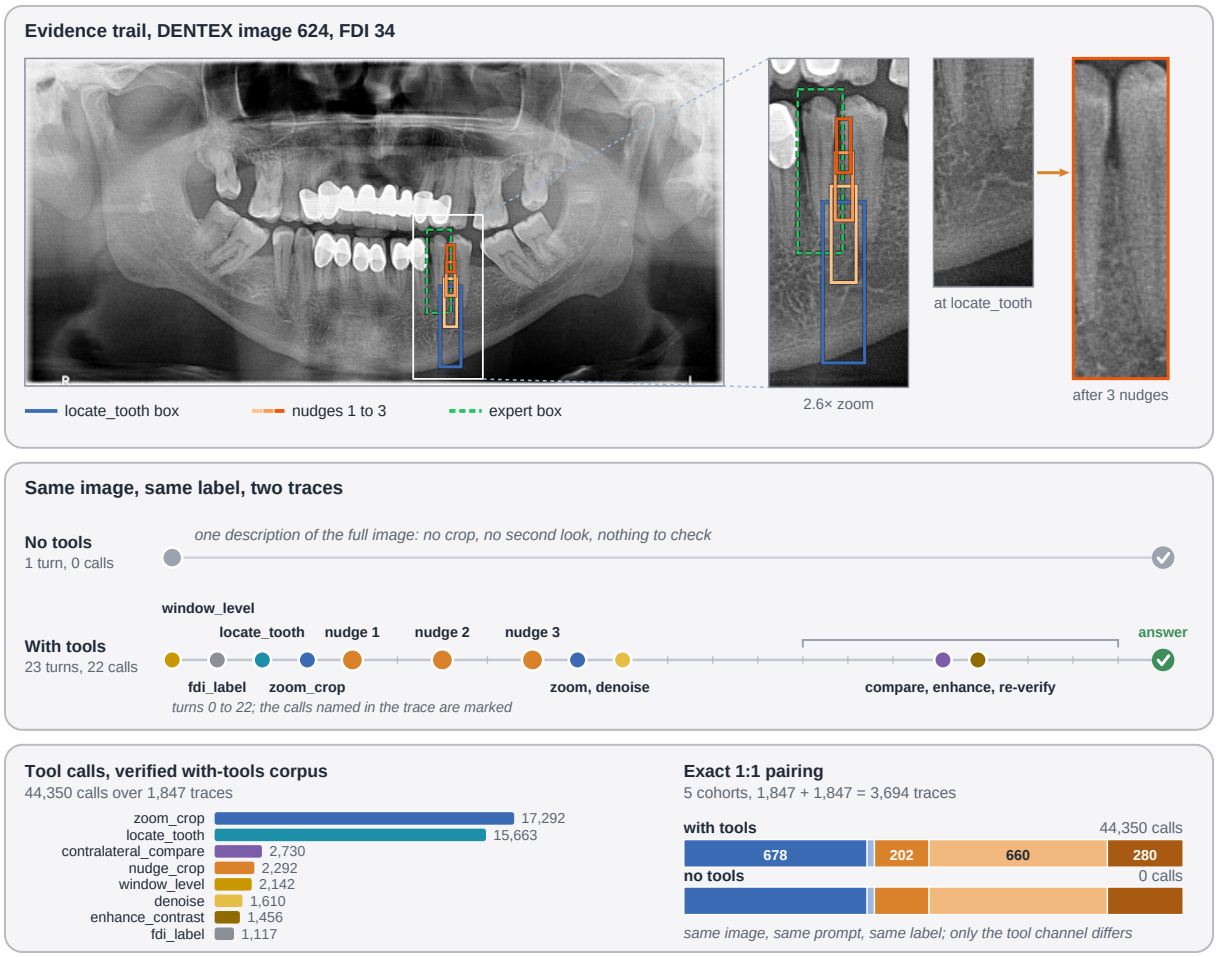


Figure 7: One DENTEX image (624, FDI 34) in both arms. Top: the real box walk. Middle: the no-tools trace (one turn) and the with-tools trace (23 turns, 22 tool calls). Bottom: tool calls in the verified with-tools corpus (44,350 attempts, including 48 invalid ones) and the exact 1:1 pairing of the two arms.

and the label fixed and vary only the tool interface, isolating what interactive tool use adds over single-pass reasoning (Section 4.1). The no-tools traces are retained deliberately: they teach output format and clinical vocabulary and provide the baseline arm of every comparison — but they are ground-truth-directed single-pass generations, closer to hindsight rationalization than to blind discovery (Section 3.6).

3.8 Training: SFT then GRPO with a correction-friendly reward

Stage 1 (SFT) fine-tunes Qwen3.5-9B (Bai et al., 2025) with QLoRA (Dettmers et al., 2023) on the verified with-tools traces (Fig. 8), teaching the *shape* of the behavior: well-formed reasoning, well-formed tool calls with continuous parameters, and structured multi-finding final answers. Multi-turn trajectories interleave the real crops the tools produced, and loss is masked to assistant tokens only.

Stage 2 (GRPO (DeepSeek-AI, 2025)) optimizes the composite reward

$$R = w_{\text{acc}} R_{\text{acc}} + w_{\text{fmt}} R_{\text{fmt}} + w_{\text{tool}} R_{\text{tool}} + w_{\text{eff}} R_{\text{eff}},$$

with three properties that matter for this task:

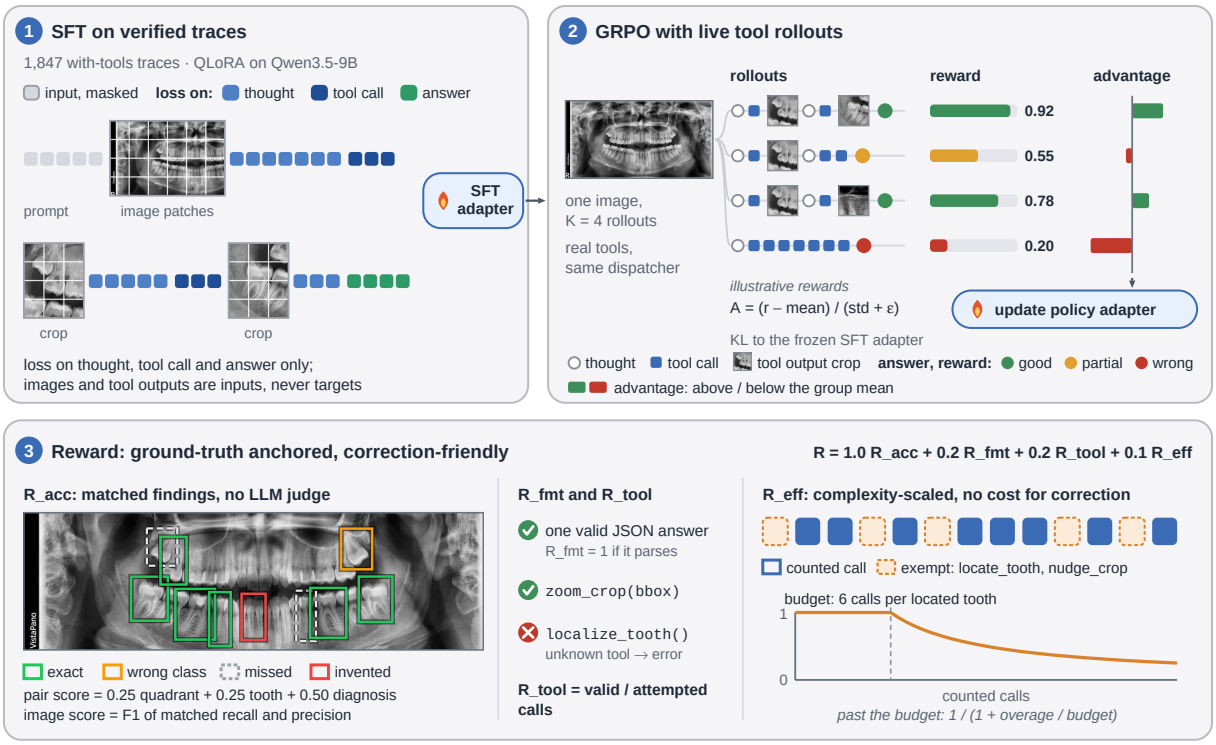


Figure 8: Training. (1) SFT on loss-masked verified traces. (2) GRPO with live tool rollouts, group size K=4, and the frozen SFT adapter as KL reference. (3) The ground-truth-anchored reward R.

Ground-truth-anchored, multi-finding accuracy. R_{acc} is a hard, non-LLM-judge match against expert labels. Predicted findings are matched to annotated findings one-to-one by greedy highest-score-first assignment; each matched pair scores $0.25 \cdot \mathbf{1}[\text{quadrant}] + 0.25 \cdot \mathbf{1}[\text{position}] + 0.50 \cdot \mathbf{1}[\text{diagnosis}]$, and the image-level score is the F1-style harmonic mean of the matched recall and precision — missed findings and hallucinated extra findings both score zero, so indiscriminate guessing cannot inflate the score. With one finding on each side this reduces exactly to the single-pair score. This design is motivated by the reward-hacking failure that Fan et al. (2026) document for an LLM-judge accuracy term. A rule-based match removes that judge dependence; it does not by itself rule out other forms of reward exploitation.

Complexity-scaled efficiency that does not fight correction. A naive per-call penalty prices every tool call equally — and therefore prices a legitimate corrective `nudge_crop` the same as an aimless call, directly fighting the behavior Section 3.5 exists to teach. R_{eff} instead derives a reference budget from the trajectory itself (6 calls per distinct located tooth), exempts `locate_tooth` and `nudge_crop` entirely from per-call cost, and charges only budget overruns and exact back-to-back repeat calls (which yield no new information), with a smooth drop-off rather than a hard cliff.

Dual-adapter RL. The frozen SFT adapter serves as the KL reference policy on the same base model, computing the KL penalty by adapter swapping rather than a second model in memory.

Table 3: Zero-shot frontier VLMs on the DENTEX validation set ($N = 50$ panoramic radiographs; set-level matching over annotated findings). Exact hit is the percentage of evaluated images with at least one correct tooth–diagnosis pair, not complete-case accuracy. FDI hit similarly requires at least one matched tooth. The reported confidence interval refers to exact hit; F1 pools finding counts. N is the available evaluated count per model, which differs across rows; these are not identical paired cohorts. ECE = expected calibration error.

Model	N	FDI hit	Exact hit	Exact F1	95% CI	ECE
Moonshot Kimi-k3	50	64.0%	**46.0%**	**0.222**	[32.0, 60.0]	**0.214**
Meta Llama 3.2 90B	46	45.7%	37.0%	0.098	[23.9, 52.2]	0.395
Gemini 3.5 Flash	50	62.0%	36.0%	0.188	[22.0, 50.0]	0.560
MiniMax M3	50	62.0%	32.0%	0.110	[20.0, 44.0]	0.493
Gemini 3.5 Flash Lite	50	54.0%	32.0%	0.166	[18.0, 46.0]	0.596
Qwen3.5-9B (our backbone)	49	53.1%	30.6%	0.112	[18.4, 42.9]	0.600
Majority-class floor	50	38.0%	8.0%	0.038	[2.0, 16.0]	–
Meta Llama 3.2 11B	50	26.0%	4.0%	0.010	[0.0, 10.0]	0.678

4 Experiments

4.1 Evaluation protocol

Data and splits. We evaluate on DENTEX's official 50-image validation set (46 pathological images with 182 annotated target teeth, plus 4 healthy controls) as the test set, used exactly once for final numbers. All configuration choices — checkpoint selection, ablation conditions, group-size sweeps — are made on a fixed dev split of ∼70 training images drawn once with a fixed seed, so that no choice leaks into the test result. Healthy images are evaluated with ground truth $\emptyset$: an invented finding on a healthy image is charged to precision. Cross-dataset context comes from the Tufts Dental Database (Tufts University School of Dental Medicine, 2023), which contributes training traces and grounding training data but is not used for test-time evaluation in this work.

Metrics. Findings are matched as a *set*: predictions and annotations are paired one-to-one by optimal assignment (Kuhn, 1955), and a finding is correct at two levels (exact: right tooth and diagnosis; tooth-only: right tooth). Counts are pooled over all findings before computing precision/recall/F1, so a healthy image's invented finding costs precision and images with many findings are not diluted by averaging per-image scores. Uncertainty is quantified by resampling whole *images* (not findings) with replacement; comparisons between conditions are paired, using the same resamples on the images the conditions share. Alongside accuracy we report the fraction of healthy images on which findings were invented, the fraction of episodes ending without a parseable answer, and expected calibration error (ECE) (Guo et al., 2017) between stated confidence and correctness.

Conditions. Every condition is a checkpoint (base, SFT, GRPO) with or without tool access, evaluated on the original images; tools always operate on the native image. The no-tools arms use the *same* SFT and GRPO recipe on the 1,847 single-turn traces, so tool contribution is not confounded with training recipe. Fig. 9 summarizes the design.

4.2 Zero-shot baselines: the case for looking again

Table 3 motivates additional inspection without identifying a causal error mechanism. The best pooled exact F1 is 0.222; the 46.0% image-level exact hit rate answers a different question and is not

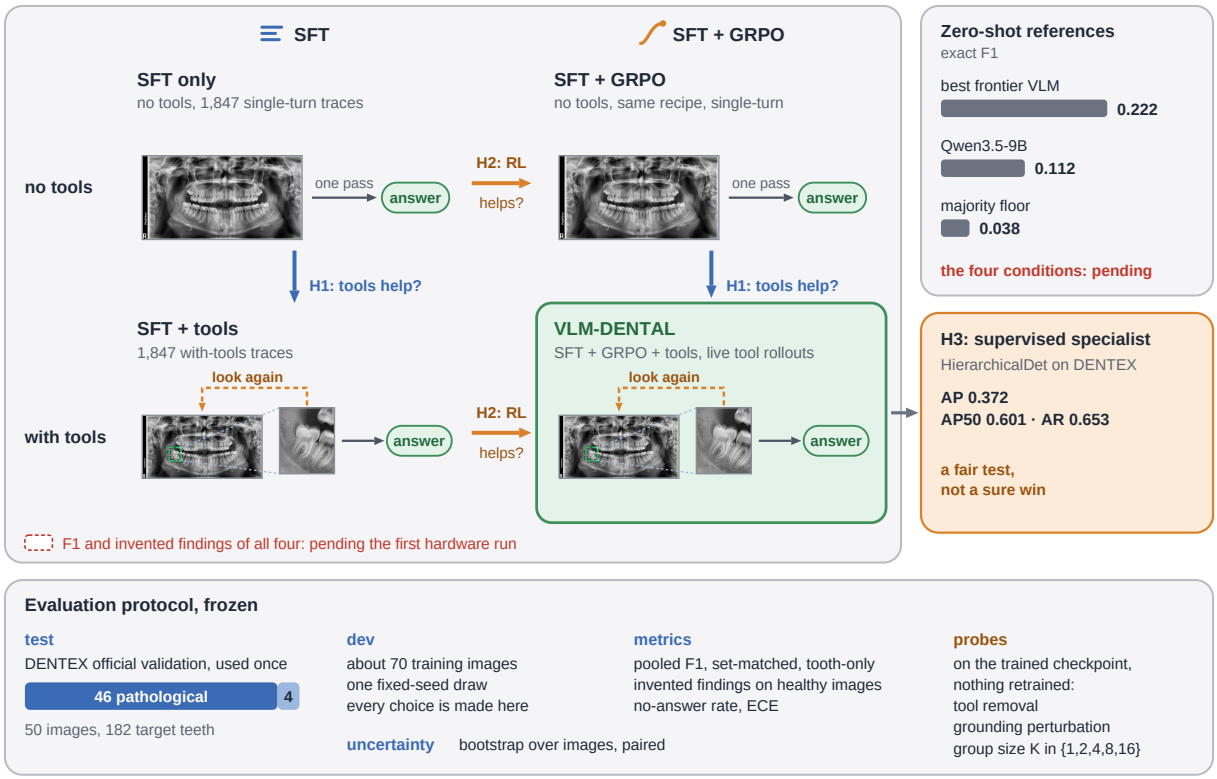


Figure 9: Experimental design: SFT or SFT+GRPO, with or without tools. H1 compares rows, H2 columns, and H3 the full system with a supervised specialist, next to the zero-shot references of Table 3. Result cells are pending the first hardware run, as in Table 4; the protocol is frozen (Section 4.1).

Table 4: Main results on the DENTEX validation set (pooled finding-level, set-matched); "Best zero-shot frontier" is the best row of Table 3. Cells marked [] are pending the first hardware training run; the experimental design and metric definitions are final and frozen (Section 4.1).

Condition	Tools	Exact F1 ↑	Tooth-only F1 ↑	Invented↓
Best zero-shot frontier	–	0.222	–	–
Zero-shot Qwen3.5-9B	–	0.112	–	–
Majority-class floor	–	0.038	–	–
SFT only (no tools)	–	[pending]	[pending]	[pending]
SFT + tools	✓	[pending]	[pending]	[pending]
SFT + GRPO (no tools)	–	[pending]	[pending]	[pending]
VLM-DENTAL (SFT + GRPO)	✓	[pending]	[pending]	[pending]
Supervised specialist (HierarchicalDet), DENTEX: AP 0.372 / AP_{50} 0.601 / AR 0.653 (Hamamci et al., 2024)				

finding recall or complete-case correctness. Tool-specific benefits require the controlled trained-agent comparisons, not extrapolation from baselines.

4.3 Main results

Table 4 is the paper's primary claim table: does the agentic recipe beat zero-shot inference and the same backbone without tools (H1: tools help; H2: RL helps), and where does it stand against a supervised specialist (H3, framed as a fair test rather than a guaranteed win). Cells for the trained conditions are pending and will be filled from the frozen evaluation protocol above.

Table 5: Correction behavior by perturbation tier, over the ground-truth-anchored subset of locate returns (reference box = expert box; pathology cohorts). P(nudge) counts any attributed `nudge_crop`; P(corrective) counts locates with at least one corrective-direction nudge; IoU before and after are paired, both over the corrected locates (the same episodes before and after their nudges). The clean tier is a control: with a correct box, corrective nudges are rare.

Tier	n	P(nudge)	P(corrective)	IoU before	IoU after	ΔIoU
none	2,338	19.4%	1.1%	1.00	0.27 ± 0.18	-0.73
small	1,337	23.1%	13.6%	0.45	0.45 ± 0.18	+0.00
big	1,493	47.2%	38.4%	0.09	0.24 ± 0.18	+0.16

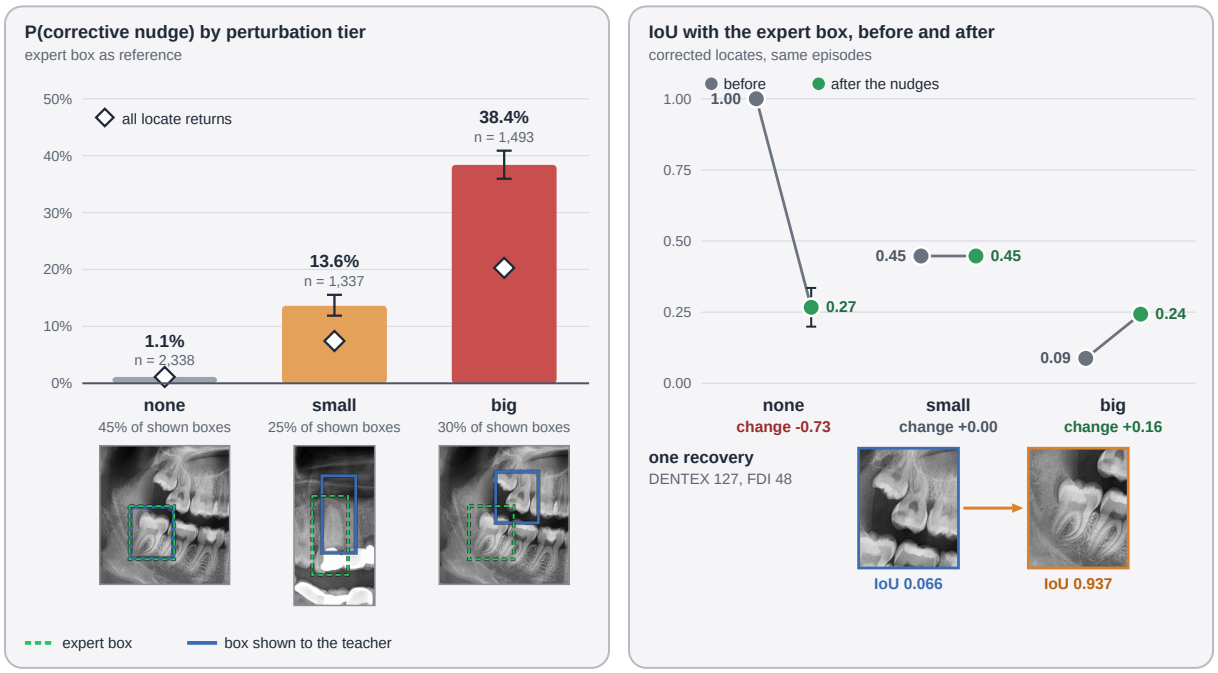


Figure 10: Correction behaviour by perturbation tier (rates in Table 5). (a) Corrective `nudge_crop` rate on the ground-truth-anchored subset (Wilson 95% intervals); diamonds: all locate returns. (b) IoU with the expert box before and after correction, same episodes (95% intervals of the mean). The few nudges on clean boxes are mostly reframing rescales.

4.4 Mechanism analysis: does tiered perturbation elicit self-correction?

Before any training, the corpus itself tests the central design claim of Section 3.5: did the perturbation hook make the *teacher* — a frontier model with no verbal hints — actually inspect, judge, and correct grounding output, rather than nudge mechanically? We answer this by re-walking every with-tools trace (8,404 analysed `locate_tooth` returns) and reconstructing, for each located tooth, the box the teacher was shown, the reference box, and every subsequent `nudge_crop` applied to that box through the trace. The shown box is recovered from the retained observation payloads (exact post-perturbation boxes; validated at zero mean coordinate error against reference boxes on the clean tier) or, where context truncation removed the observation, from the teacher's own downstream `bbox` arguments, attributed by nearest center. Each attributed nudge is then classified by direction: *corrective* if its shift moved the working box toward the reference center (cosine > 0.3), *away* if against it, *rescaling* otherwise.

Table 5 and Fig. 10 show the intended monotonic gradient: corrective nudges on 38.4% of large offsets, 13.6% of small ones, and 1.1% of clean boxes. The clean-tier control matters: the teacher does not nudge reflexively — with a correct box it proceeds 98.9% of the time, so the gradient is driven by the *observed* displacement, not a scripted habit. The small tier behaves as designed, as a judgment call: nudges there are frequent (23.1%) but only sometimes corrective (13.6%), consistent with the zoom margin often absorbing a small offset (the paired mean IoU of those episodes is unchanged, 0.45 → 0.45). On corrected large offsets, mean IoU against the expert box improves from 0.088 to 0.243 on the same 573 episodes — partial on average, because a single nudge rarely lands perfectly — with individual recoveries as large as IoU 0.066 → 0.937 on a DENTEX training image (Fig. 1; further episodes in Fig. 11). The remaining gap is visible in the audit: uncorrected spatial drifts are what the verifier's third rejection category (11 traces) caught and repaired during corpus construction.

Two further observations qualify these results. First, the split by reference type (full split in Appendix B): on *detector-anchored* locates (exploratory calls during healthy-scan full-arch surveys, where the reference is the detector's own box), correction is much rarer (5.0% on large offsets vs. 38.4% ground-truth-anchored); during rapid systematic surveys the teacher surveys broadly and inspects each exploratory box less carefully. The demonstrated skill this paper targets is the ground-truth-anchored pathology workflow. Second, nudge *rescales* skew small (scale < 1): the dominant corrective pattern is tighten-after-shift, which resembles zooming into a suspected lesion after centering it.

4.5 Ablations and robustness checks

The following probes run on a trained checkpoint without retraining; their results are pending.

Tool-removal probe. A tool is removed from the registry (absent from the prompt's tool list; calls to it return the standard "not recognized" error), and we record both the score drop and how often the trained agent attempts the missing tool. This measures dependence on each tool, including `nudge_crop`; because the checkpoint was trained with the tools present, it is a dependence probe, not the score of a tool-free model; the no-tools training arms in Table 4 serve the latter role. **[pending]**

Grounding-perturbation probe. Every `locate_tooth` box at evaluation time is displaced by a fixed pseudo-random fraction of its own size (the same wrong box for the same image and tooth across conditions, so comparisons stay paired), and we report the IoU of the agent's working box against the annotated tooth before and after its last `nudge_crop`, plus how much of the annotated tooth the last `zoom_crop` actually shows. This is a stress test of the trained agent's self-correction — the deployed version of Section 4.4. **[pending]**

Group-size sweep. GRPO group sizes $K \in \{1, 2, 4, 8, 16\}$ on the dev split, reporting pooled F1 per K with paired comparisons and training cost. The sweep is descriptive and is not used for selection. **[pending]**

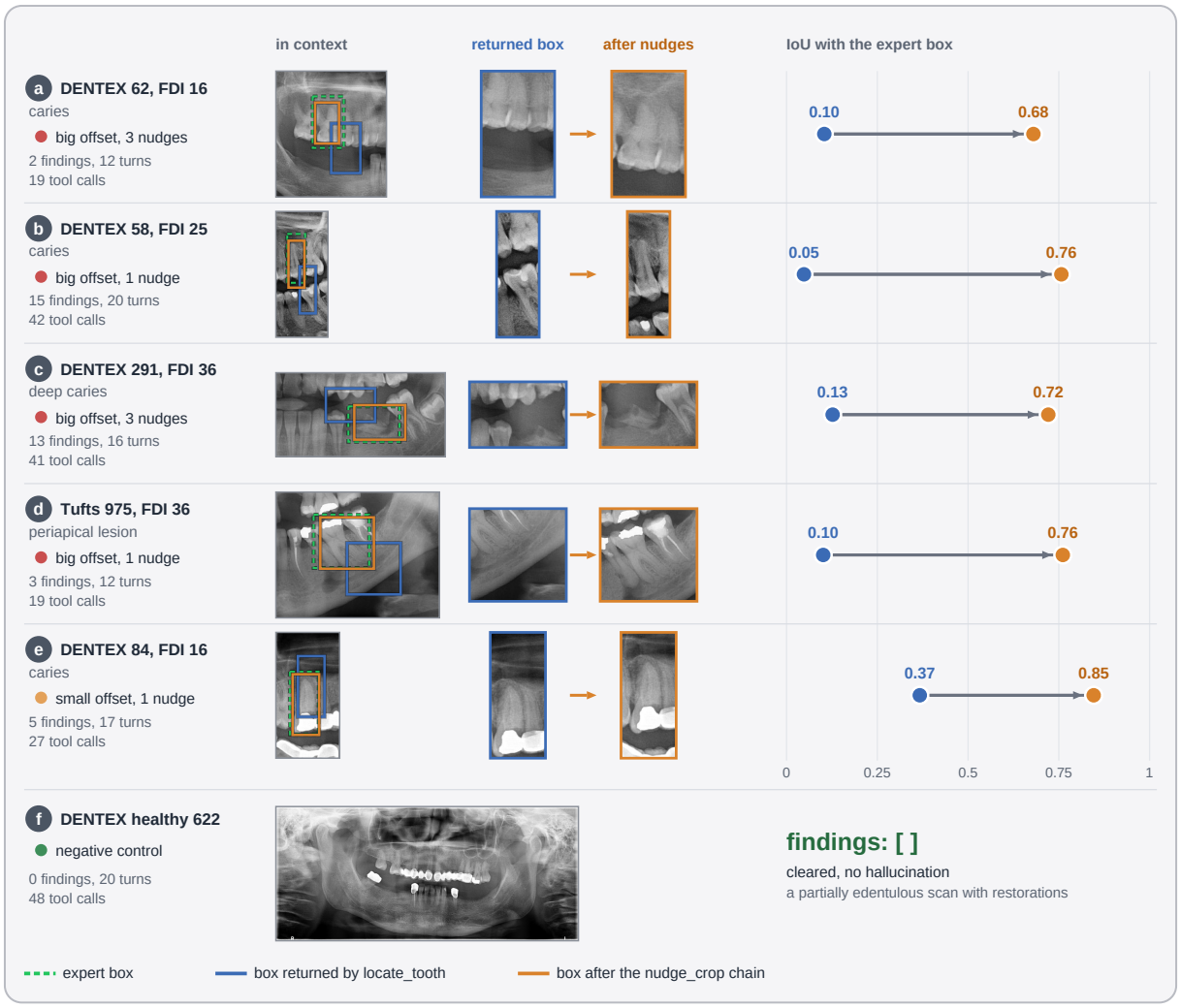


Figure 11: Six verified episodes (five pathology, one healthy control) from two institutions: expert box green, box returned by `locate_tooth` blue, box after the `nudge_crop` chain orange, with the crops the agent inspected before and after. (a) Three nudges; (b) one nudge among 15 findings; (c) deep caries; (d) Tufts periapical lesion; (e) small offset; (f) healthy scan cleared without a hallucinated finding. Tiers are the logged perturbation tiers.

5 Inference harness and serving: a separate dev-only extension

This extension follows checkpoint training. It changes neither the SFT/GRPO objectives nor the frozen primary evaluation in Section 4.1. All effectiveness and speed results below are pending. Its purpose is to distinguish three possible sources of a change in behavior: learned weights, procedural context at inference, and decoding implementation.

5.1 Versioned rules and progressive procedural context

Figure 2 (Extension 1) places the harness in the agent loop; it supplies procedural context and audits its use. The harness contract extracts existing system/tool instructions and verifier/reward constraints into `rules.md`. Versioned rule identifiers link statements to prompt rendering or enforcement checks. A byte-level test requires the default file-based rendering to reproduce the existing

Table 6: Exploratory, paired dev-only harness comparison on one frozen checkpoint. Finding invention is $FP/(TP+FP)$, distinct from the primary healthy-image metric. Pending cells are not measurements.

Context condition	Exact F1	Finding invention
Baseline	[pending]	[pending]
Rules only (identity control)	[pending]	[pending]
Rules + on-demand skills	[pending]	[pending]
Rules + full skills	[pending]	[pending]

system prompt exactly; inventory checks bind verifier/reward identifiers to their implementations without changing outcomes or reward values. Code-only and prompt-only constraints are recorded as discrepancies, not silently reconciled.

Five versioned procedural skills cover box verification, caries versus deep caries, periapical lesions, impacted teeth, and a healthy-scan negative survey. Sources accompany each skill; content is drawn from existing procedures, class definitions, and cited public material, without test images or test-derived examples. Skills are contextual instructions, not new learned parameters. With skills disabled, the existing agent is unchanged. In `index_only` mode, the prompt lists names and descriptions, and a `read_skill` harness action loads a body on demand. In `full_inject` mode, all bodies are present initially. The harness action is dispatched outside the unchanged eight-tool diagnostic registry; it does not consume a diagnostic-call slot, but the episode turn bound still applies. Loaded skill versions, content digests, and turn/timing events are recorded in optional trace metadata. Old traces remain readable without that metadata.

5.2 Paired evaluation without reopening the primary protocol

The extension (Table 6) uses only the fixed, training-derived dev split. A frozen checkpoint is evaluated under baseline, rules only, rules plus index/on-demand skills, and rules plus full skills. The same image identifiers, seeds, geometry, dispatcher, tool budgets, and scoring implementation are used across conditions; full skills are a disclosure control, not an additional training arm. Rules-only and baseline must render identical prompts; any deviation invalidates the control. We report pooled exact F1 and the finding-level false-discovery fraction

$$F_1^{\text{exact}} = \frac{2TP}{2TP + FP + FN}, \qquad I_{\text{finding}} = \frac{FP}{TP + FP}.$$

The latter is the harness JSON field `invented_findings_rate`; it is *not* the healthy-image invention rate of Section 4.1. A zero denominator is undefined, not zero. Paired uncertainty resamples whole shared images. We also report format failures and disclosure events alongside the primary endpoints. Because these conditions were added after the primary protocol was frozen, they are exploratory and do not amend that protocol. Dev-split selection cannot establish independent generalization.

5.3 Speculative decoding: accuracy and end-to-end efficiency

Speculative decoding accelerates autoregressive generation by proposing several next tokens and checking them together with the target model (Leviathan et al., 2023; Chen et al., 2023). Accepted proposals extend the output without the same number of sequential target-decoding steps. Its role in this system is confined to the local serving backend (Fig. 2, Extension 2, and the MTP block of Fig. 3): the agent still consumes the image and conversation, produces a structured turn, invokes the shared dispatcher, and receives real tool observations. Image prefill and diagnostic-tool execution remain separate costs; proposal overhead and rejected tokens can offset decoding gains.

Table 7: Accuracy and efficiency of speculative decoding on paired dev episodes. Diagnostic accuracy is pooled exact F1. Speedup is baseline total episode wall time divided by speculative total wall time. Proposal hit rate is accepted proposed tokens divided by all proposed tokens, not diagnostic accuracy. The baseline has no proposals. Failed decoded-output or F1 agreement withholds speedup; missing measurements remain pending.

Decoding	Exact F1	Speedup	Hit rate	Agreement
Baseline	**[pending]**	**[pending]**	–	pending
Speculative	**[pending]**	**[pending]**	**[pending]**	pending

The comparison uses one merged, trained checkpoint with speculative decoding disabled or enabled. Both conditions use the same fixed dev images, prompt construction, greedy decoding, seeds, tool implementation and budgets. The serving configuration, hardware, software versions and checkpoint identity accompany the result artifact. When the selected serving configuration uses the native MTP head, the checkpoint audit checks its tensors against the original base before execution.

The planned draft-training stage uses with-tools training contexts and continuations generated by the trained target. The target remains frozen while draft parameters learn to predict its continuations. Dev and test images do not enter this stage. The resulting draft supplies proposals in Fig. 2; the target continues to verify them. Acceptance and end-to-end efficiency require measurements after training; no completed draft-training result is claimed here.

Table 7 reports diagnostic accuracy as pooled exact F1, end-to-end speedup as the baseline's total episode wall time divided by the speculative condition's total wall time, and proposal hit rate as accepted proposed tokens divided by all proposed tokens. The hit rate is a decoding statistic, not diagnostic accuracy; it is undefined for the baseline. Timing includes the complete agent episode so that a decoding improvement is not mistaken for the same proportional reduction in image-processing or tool time. Separately exposed prefill, decode and tool counters remain in the result artifact.

Decoded turns, tool actions and final answers are compared on each shared image, together with pooled exact F1. Divergences remain visible and disqualify an equivalence-qualified speedup; an aggregate score match alone is insufficient. This finite dev comparison does not establish token-ID equivalence or equivalence under stochastic sampling. The manuscript table contains only the explicitly selected speculative configuration, recorded with its provenance, and the paired baseline; it does not select the fastest setting from the evaluation results. All unmeasured entries remain pending.

6 Limitations

No prospective clinician study. Our validity anchor is DENTEX's dentist-verified ground truth (each image was annotated by a final-year dental student and verified by an expert dentist with over 15 years of experience), together with automated checks of reasoning grounding by a cross-family LLM judge. This validates against expert labels, not against a prospective, blinded review by practicing clinicians on new cases. A reader study is the natural next step.

Pending trained-agent results. The training-side evaluations (SFT, GRPO, and ablations on the trained agent) are designed, frozen, and implemented, but the runs are not complete. Those cells are marked as pending instead of extrapolated. The completed evidence (zero-shot baselines, grounding-tool validation, and the corpus-level mechanism analysis) stands independently, but end-to-end claims cannot be made until the runs are complete.

Synthetic perturbations are not real detector errors. The tiered perturbation draws offsets from fixed configured ranges, not from the grounding model's actual error distribution. This keeps demonstrations stable as the detector improves (Section 3.5), but the training distribution of box errors is synthetic: uniform random offsets with random signs, unlike the structured errors a real detector makes (for example adjacent-tooth confusions and root-tip truncations). The evaluation-time grounding-perturbation probe (Section 4.5) measures robustness to a related synthetic distribution. Robustness to the detector's true error modes is best assessed by swapping in improved detector weights, which our decoupling makes safe for the trace corpus but which does not replace measuring it.

Demonstrated vs. learned correction. Section 4.4 measures the *teacher's* behavior in the demonstrations, which is the skill the student is trained to acquire, not the trained agent's behavior at inference. The agent-side version of the same analysis is the grounding-perturbation probe, which awaits the trained checkpoint. Demonstrated behavior is necessary but not sufficient for learned behavior.

Scale and scope. The corpus holds 3,694 traces over two institutions and four DENTEX diagnosis classes (plus Tufts' four-class taxonomy). Prior work suggests that a few hundred verified examples can suffice for medical RL fine-tuning (Hu et al., 2025), but the corpus is modest in size and the DENTEX test set contains only 50 images, so confidence intervals on it are wide; we report them rather than over-reading point estimates. The tool suite covers radiologist-like inspection primitives. Measurement and calibration tools (lesion sizing, periodontal probing analogues) and anomaly detection beyond the four classes are out of scope.

Automated verification is not expert review. The verifier is a frontier model from a different family than the generator, audited against expert labels, and its rejection taxonomy shows that it catches real failures. Automated verification nevertheless has blind spots that human clinical review would not. We mitigate this with cross-family verification, ground-truth anchoring of every correctness claim, and release of the full corpus and audit trail so that others can re-audit it.

Harness and serving evidence. The rules-only condition is a prompt-identity control, not a proposed source of diagnostic improvement. Skills add procedural context and may help or distract, so their effects require paired dev-split measurements. A BF16 merged target is numerically different from a 4-bit QLoRA serving target, so speed or equivalence cannot be attributed to speculation across those targets. Multimodal and MTP support and memory feasibility must be established for the pinned runtime and hardware; a recognized configuration flag is not evidence of a completed run. Neither the harness nor speculative decoding establishes clinical validity.

7 Conclusion

VLM-DENTAL makes grounding correction an explicit action in a dental VLM pipeline. Box-relative revision, deliberately imperfect demonstration-time grounding, and a correction-friendly reward together define a testable approach to learning when to look again. The completed corpus analysis shows a teacher correction gradient across perturbation tiers; it does not show that the trained student corrects errors or improves diagnostic accuracy. Those conclusions require the pending SFT/GRPO comparisons under the frozen protocol.

The inference extension separates an exact rules rendering from on-demand procedural skills, and the serving extension tests speculative decoding against the same numerical target. Their quality and efficiency results are reported independently, with missing measurements left explicit. Together with the released sources and audit trail, this separation keeps both positive and negative results interpretable. The next step is to complete the predefined trained-agent comparisons, followed by broader external validation.

Data and code availability

The verified trace corpus, the grounding model weights, and model checkpoints are hosted on the Hugging Face Hub. **[Add repository, Hub, and DOI links before submission.]**

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Table 8: Exact tiered-perturbation configuration used for trace generation. Offsets are drawn uniformly per axis with a random sign and applied with `nudge_crop`'s own shift-and-clamp math, so every shown box is one the correction tool could produce or fix. The unperturbed box is logged for auditing and never shown to the model.

Tier	Probability	Offset magnitude (per axis, box-relative)	Intent
none	0.45	0	Clean accept-and-proceed demonstration
small	0.25	$U(0.12, 0.28)$, random sign	Genuine judgment call; not resolved by the prompt
big	0.30	$U(0.45, 0.75)$, random sign	Visually self-evident without any verbal cue

A Perturbation configuration

The exact configuration used for trace generation is given in Table 8.

Rates are fixed constants, deliberately decoupled from the grounding detector's measured accuracy so that improving the detector never invalidates existing demonstrations (Section 3.5).

B Mechanism analysis: full split by reference type

Table 9 reports the correction-behavior analysis split by reference type; Section 4.4 discusses the ground-truth-anchored rows in the main text.

Table 9: Corrective-nudge behavior split by reference type. *GT-anchored*: the locate call was ground-truth-grounded (annotated tooth; pathology cohorts) and the reference box is the expert box. *Detector-anchored*: exploratory calls (dominated by healthy-scan full-arch surveys), where the reference is the detector's own box. During rapid surveys the teacher corrects far less often per box, a difference in behavior that pooled rates would hide.

Reference	Tier	n	P(nudge)	P(corrective)	IoU before → after
GT	none	2,338	19.4%	1.1%	1.00 → –
GT	small	1,337	23.1%	13.6%	0.46 → 0.45
GT	big	1,493	47.2%	38.4%	0.10 → 0.24
Detector	small	1,472	2.9%	1.8%	0.37 → 0.38
Detector	big	1,764	6.9%	5.0%	0.09 → 0.15

Recovery rates of the shown box (from retained observations or downstream `bbox` arguments) were 0.97–1.00 in every cell; validation of observation-derived boxes against reference boxes on the clean tier gave zero mean coordinate error.

C Grounding tool validation

Table 10 summarizes the grounding tool's validation.

The precision/recall profile is the intended operating point for a grounding *tool* inside an agent loop: high precision so crops are plausible, non-perfect recall so that correction (Section 3.4) and re-location remain useful.

Table 10: YOLOv8m grounding tool (32 FDI classes), 5-fold cross-validation on the combined DENTEX+Tufts pool (2,339 images, 46,808 boxes) and evaluation on the held-out target benchmark (50 images, 182 annotated targets, greedy one-to-one target matching). "Target mAP50" restricts evaluation to detections on annotated target teeth.

Model	Target mAP50	Target mAP50-95	Precision	Recall@0.50
DENTEX-only (5-fold mean)	0.909	0.638	0.939	0.818
DENTEX+Tufts (5-fold mean)	0.926	0.644	0.963	0.775
Best fold (DENTEX+Tufts, fold 1)	**0.959**	**0.650**	**0.986**	0.797

D Verifier rejection taxonomy

During first-pass verification the independent cross-family verifier rejected 92 of 3,694 traces (97.5% first-pass rate), all subsequently repaired and re-verified to a 100% final yield:

1. **False-positive pathology on healthy scans** — 66 rejections (71.7%). The generator hallucinated findings on disease-free scans; repaired to systematic negative surveys committing to `[]`.
2. **Ground-truth contradictions and FDI errors** — 15 rejections (16.3%). Mismatched pathology classes or swapped quadrants; repaired via targeted re-prompting.
3. **Uncorrected spatial drift** — 11 rejections (12.0%). The teacher diagnosed from a visibly displaced grounding box without a corrective `nudge_crop`; repaired with spatial realignment. This category exists because of the perturbation mechanism (Section 3.5) and indicates that it creates judgment pressure the verifier can check.

In a separate decontamination pass, 105 traces in which the generator echoed its hidden ground-truth directive in its reasoning were excised and regenerated under a dual gate (zero-leak audit + verifier re-approval); the canonical corpus contains zero directive leaks and exact 1:1 parity between the with-tools and no-tools cohorts.